VCE GENERAL MATHEMATICS · PRACTICE EXAM

General Mathematics · Harder paper

Real VCAA questions ranked by the published state results. Editable: type your answers straight into this PDF, save it, then mark yourself with the guide at the back.

Name _____ **Date** _____

Exam 1 · multiple choice	Exam 2 · written response	State average on these questions
40 marks	**59 marks**	**43 / 99 (44%)**
reading 15 min · writing 1 h 30 min	reading 15 min · writing 1 h 30 min	a student at the state average on every question

How to use this paper

1. Open each question with its blue link. It opens the official VCAA exam PDF at the right page. The question text isn't reproduced here because it is VCAA copyright.
2. Answer in this file: click a bubble for multiple choice, and type in the boxes for written answers (long answers scroll inside the box). Working can be typed in plain text, e.g. 0.004 × 5440.88 = 21.76.
3. Keep to the real time limits above, with your CAS, your bound reference and the VCAA formula sheet, and don't look at the guide until you've finished.
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5. Mark yourself: in the guide at the back, tick each marking point you earned, enter your marks, and compare your total with the state average.

Why these questions

Every question is a real VCAA exam question from the 2023–2025 General Mathematics exams, chosen by ranking all the questions by how Victorian students actually scored on them (percentage correct for multiple choice, state average ÷ marks for written questions) within each area of study. This paper takes the lowest band, with the real exam's balance of areas.

Sources

- 2023: Exam 1 (PDF) · report · Exam 2 (PDF) · report
- 2024: Exam 1 (PDF) · report · Exam 2 (PDF) · report
- 2025: Exam 1 (PDF) · report · Exam 2 (PDF) · report

Exam 1 · multiple choice (40 marks, reading 15 min, writing 1 h 30 min)

Open each question in the official exam PDF, then click your answer. 2023 questions have five options (A–E); 2024 and 2025 questions have four (A–D).

Data analysis

Question 1 · 1 mark

2023 VCAA Exam 1 · Question 6 · open in the official exam PDF (p. 5) ↗

How many of six car prices fall in the modal class of a $\log_{10}$(price) histogram

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

Question 2 · 1 mark

2023 VCAA Exam 1 · Question 7 · open in the official exam PDF (p. 6) ↗

Least squares line for test 2 against test 1 marks from a scatterplot

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

Question 3 · 1 mark

2023 VCAA Exam 1 · Question 8 · open in the official exam PDF (p. 6) ↗

Count students whose test 2 mark is within two marks of the line's prediction

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

Question 4 · 1 mark

2023 VCAA Exam 1 · Question 9 · open in the official exam PDF (p. 7) ↗

Slope of a least squares line from two predicted points (food energy and birth rate)

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

Question 5 · 1 mark

2023 VCAA Exam 1 · Question 10 · open in the official exam PDF (p. 7) ↗

What a negative association ($r^2 = 0.72$) between social media time and test scores allows you to conclude

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

Question 6 · 1 mark

2023 VCAA Exam 1 · Question 11 · open in the official exam PDF (p. 8) ↗

Least squares line for tree height against 1/age

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

Question 7 · 1 mark

2023 VCAA Exam 1 · Question 14 · open in the official exam PDF (p. 10) ↗

Median winning time over all 23 years from the time series plot

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

Question 8 · 1 mark

2023 VCAA Exam 1 · Question 16 · open in the official exam PDF (p. 11) ↗

Seasonal index for a month whose figures are increased by 35% when deseasonalised

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

Question 9 · 1 mark

2024 VCAA Exam 1 · Question 6 · open in the official exam PDF (p. 5) ↗

Smallest athlete count that would be an outlier, from a five-number summary

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

Question 10 · 1 mark

2024 VCAA Exam 1 · Question 9 · open in the official exam PDF (p. 7) ↗

Correlation coefficient from the slope and summary statistics

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

Question 11 · 1 mark

2024 VCAA Exam 1 · Question 11 · open in the official exam PDF (p. 8) ↗

Least squares line after a $\log_{10}$ transformation of the explanatory variable (year)

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

Question 12 · 1 mark

2024 VCAA Exam 1 · Question 12 · open in the official exam PDF (p. 8) ↗

Year with the largest residual after a reciprocal transformation of pairs

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

Question 13 · 1 mark

2024 VCAA Exam 1 · Question 16 · open in the official exam PDF (p. 10) ↗

Missing seasonal index for month 3, given it is twice month 6

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

Question 14 · 1 mark

2025 VCAA Exam 1 · Question 4 · open in the official exam PDF (p. 4) ↗

Which column of a $\log_{10}$(population density) histogram contains Singapore

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

Question 15 · 1 mark

2025 VCAA Exam 1 · Question 12 · open in the official exam PDF (p. 7) ↗

Prediction from a life = a + b × doctors² model for 2 doctors per 1000 people

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

Question 16 · 1 mark

2025 VCAA Exam 1 · Question 16 · open in the official exam PDF (p. 9) ↗

Percentage reduction to deseasonalise winter pie sales with index 1.75

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

Recursion and financial modelling

Question 17 · 1 mark

2023 VCAA Exam 1 · Question 17 · open in the official exam PDF (p. 12) ↗

Second term of a recurrence relation that changes the sign of each term

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

Question 18 · 1 mark

2023 VCAA Exam 1 · Question 22 · open in the official exam PDF (p. 13) ↗

Final repayment that exactly clears a $500 000 loan given by a recurrence relation

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

Question 19 · 1 mark

2023 VCAA Exam 1 · Question 23 · open in the official exam PDF (p. 14) ↗

Effective interest rate from a graph of a quarterly loan balance

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

Question 20 · 1 mark

2023 VCAA Exam 1 · Question 24 · open in the official exam PDF (p. 14) ↗

Expression for R in a perpetuity recurrence relation

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

Question 21 · 1 mark

2024 VCAA Exam 1 · Question 17 · open in the official exam PDF (p. 11) ↗

Values of R and d that generate a geometric sequence

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

Question 22 · 1 mark

2024 VCAA Exam 1 · Question 20 · open in the official exam PDF (p. 12) ↗

Simple interest rate that earns the same as a quarterly-compounding investment over three years

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

Question 23 · 1 mark

2024 VCAA Exam 1 · Question 24 · open in the official exam PDF (p. 14) ↗

New monthly addition needed after two years for an investment to reach $30 000

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

Question 24 · 1 mark

2025 VCAA Exam 1 · Question 23 · open in the official exam PDF (p. 12) ↗

Interest earned in five years given a fortnightly-compounding effective rate of 4.51%

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

Matrices

Question 25 · 1 mark

2023 VCAA Exam 1 · Question 27 · open in the official exam PDF (p. 16) ↗

Long-term behaviour of birds moving between three locations

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

Question 26 · 1 mark

2023 VCAA Exam 1 · Question 29 · open in the official exam PDF (p. 18) ↗

Build a 3 × 2 matrix from the rule $k_{ij} = (i - j)^2$

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

Question 27 · 1 mark

2023 VCAA Exam 1 · Question 30 · open in the official exam PDF (p. 18) ↗

How many of four statements about inverses and determinants are true

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

Question 28 · 1 mark

2023 VCAA Exam 1 · Question 32 · open in the official exam PDF (p. 19) ↗

From an incomplete transition diagram, the percentage of Wednesday's oval students who were also at the oval on Tuesday

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

Question 29 · 1 mark

2024 VCAA Exam 1 · Question 27 · open in the official exam PDF (p. 16) ↗

Value of g that makes a 2 × 2 matrix singular

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

Question 30 · 1 mark

2024 VCAA Exam 1 · Question 29 · open in the official exam PDF (p. 17) ↗

Which player has a bye in week 4 when a permutation matrix reorders players weekly

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

Question 31 · 1 mark

2024 VCAA Exam 1 · Question 31 · open in the official exam PDF (p. 19) ↗

Which ranking cannot happen after the last two games of a round robin

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

Question 32 · 1 mark

2024 VCAA Exam 1 · Question 32 · open in the official exam PDF (p. 20) ↗

Change in people at the Botanical Gardens from Thursday to Sunday, working backwards with $S_{n+1} = TS_n + A$

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

Networks and decision mathematics

Question 33 · 1 mark

2023 VCAA Exam 1 · Question 40 · open in the official exam PDF (p. 24) ↗

Which single change to the maze gives the largest increase in flow

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D** ◯ **E**

Question 34 · 1 mark

2024 VCAA Exam 1 · Question 35 · open in the official exam PDF (p. 22) ↗

Number of faces of a graph drawn with crossings

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

Question 35 · 1 mark

2024 VCAA Exam 1 · Question 37 · open in the official exam PDF (p. 24) ↗

Unknown edge weights that make the shortest path 34 m

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

Question 36 · 1 mark

2024 VCAA Exam 1 · Question 38 · open in the official exam PDF (p. 24) ↗

How many of four statements must be true for a connected graph with six vertices and six edges

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

Question 37 · 1 mark

2024 VCAA Exam 1 · Question 39 · open in the official exam PDF (p. 25) ↗

Time saved when a new worker replaces one of four in a task allocation

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

Question 38 · 1 mark

2024 VCAA Exam 1 · Question 40 · open in the official exam PDF (p. 26) ↗

Which precedence table matches an unlabelled activity network

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

Question 39 · 1 mark

2025 VCAA Exam 1 · Question 38 · open in the official exam PDF (p. 21) ↗

Shortest distance from home to school on a road network

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

Question 40 · 1 mark

2025 VCAA Exam 1 · Question 40 · open in the official exam PDF (p. 24) ↗

Float of activity B in a 12-activity project

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

Exam 2 · written response (59 marks, reading 15 min, writing 1 h 30 min)

Open each question in the official exam PDF, then type your answer to every part in the boxes. Show working where more than one mark is available.

Data analysis

Question 1 (6 marks)

2023 VCAA Exam 2 · Question 3 · open in the official exam PDF (p. 6) ↗

Ice cream consumption against temperature: least squares line, correlation, interpretation and prediction

a. Draw the given least squares line on the scatterplot (1 mark)

b. Correlation coefficient from $r^2 = 0.7212$, to three decimal places (1 mark)

c. Describe the association: strength, direction and form (1 mark)

d. Interpret the intercept (1 mark)

e. Predicted consumption at −6 °C (1 mark)

f. Is that prediction interpolation or extrapolation? (1 mark)

Question 2 (10 marks)

2024 VCAA Exam 2 · Question 3 · open in the official exam PDF (p. 6) ↗

Women's gold medal height (Wgold) and best height that year (Wbest): least squares line, residuals and reliability

a. Response variable in Wbest = 0.300 + 0.860 × Wgold (1 mark)

b. Draw the least squares line on the scatterplot (1 mark)

c. Coefficient of determination as a percentage, to one decimal place, from r = 0.9318 (1 mark)

d. Strength and direction of the association (1 mark)

e. Interpret the slope (1 mark)

f. Show the residual for 1984 (Wgold 2.02, Wbest 2.07) is 0.0328 (2 marks)

g.i. Add that residual to the residual plot (1 mark)

g.ii. Does the residual plot support a linear model? (1 mark)

h. Why a prediction for Wgold = 1.90 m isn't reliable (1 mark)

Question 3 (8 marks)

2025 VCAA Exam 2 · Question 4 · open in the official exam PDF (p. 6) ↗

Sale price against distance from the city centre: least squares line, r, prediction and residuals

a. Explanatory variable (1 mark)

b. Correlation coefficient from $r^2 = 0.0806$, to three decimal places (1 mark)

c. Predicted price for a home in the city centre (1 mark)

d. Is a prediction at 2 km interpolation or extrapolation? Explain (1 mark)

e. Strength and direction of the association (2 marks)

f.i. Show the residual for a home 15.5 km away that sold for $1 250 000 is 27 984 (1 mark)

f.ii. Plot that residual on the residual plot (1 mark)

Recursion and financial modelling

Question 4 (5 marks)

2023 VCAA Exam 2 · Question 7 · open in the official exam PDF (p. 13) ↗

A \$60 000 holiday loan compounding weekly, $V_{n+1} = 1.0015V_n - d$

a. Show the interest rate is 7.8% per annum (1 mark)

b.i. d for interest-only repayments (1 mark)

b.ii. d to repay the loan fully in five years (1 mark)

c. New d to repay the loan in three more years after a year of \$300 payments (1 mark)

d. Value of d that makes the recurrence relation geometric (1 mark)

Question 5 (2 marks)

2024 VCAA Exam 2 · Question 8 · open in the official exam PDF (p. 13) ↗

A \$500 000 loan at 5.3% p.a. monthly with \$3071.63 repayments: total cost and number of payments

Total cost of the loan to the nearest cent, and the number of payments (2 marks)

Question 6 (3 marks)

2025 VCAA Exam 2 · Question 9 · open in the official exam PDF (p. 13) ↗

A $50 000 loan with weekly repayments of $75: interest-only case, then a reducing balance

a. Weekly interest rate if the balance never changes (1 mark)

b.i. Annual rate (two decimal places) if the balance after a year is $49 565.34 (1 mark)

b.ii. Value of R to four decimal places (1 mark)

Question 7 (2 marks)

2025 VCAA Exam 2 · Question 10 · open in the official exam PDF (p. 14) ↗

A 10-year, $650 000 annuity at 6.4% compounding quarterly: recurrence relation for the second half

Complete D_0 and the payment term in $D_{n+1} = 1.016D_n + \ldots$ (2 marks)

Matrices

Question 8 (4 marks)

2024 VCAA Exam 2 · Question 12 · open in the official exam PDF (p. 19) ↗

Construction staff moving between jobs or leaving (absorbing state), then a model with new hires $V_{n+1} = RV_n + Z$

a. Percentage decrease in foremen from 2023 to 2025 (1 mark)

b. Total staff on site in the long term (1 mark)

c. Increase in staff on site from 2023 to 2024 in the new model (1 mark)

d. First year the number of foremen exceeds 200 (1 mark)

Question 9 (4 marks)

2025 VCAA Exam 2 · Question 13 · open in the official exam PDF (p. 18) ↗

Children switching between cooking, gardening and music each week (transition matrix K)

a. What the zeros on the leading diagonal mean (1 mark)

b.i. Expected percentage in cooking in week 10 if all 27 start in cooking (1 mark)

b.ii. Expected number in gardening in week 3 who move to music in week 4 (2 marks)

Question 10 (3 marks)

2025 VCAA Exam 2 · Question 14 · open in the official exam PDF (p. 20) ↗

A permutation matrix that reorders a holiday program's seven daily activities

a. Activities always held at the same time (1 mark)

b. Order of activities on day three (1 mark)

c. What $P^4 = I$ means over the 40-day program (1 mark)

Networks and decision mathematics

Question 11 (5 marks)

2023 VCAA Exam 2 · Question 14 · open in the official exam PDF (p. 24) ↗

A 12-activity renovation project network with a dummy, plus crashing within a budget

a. Immediate predecessors of activity I (1 mark)

b. Earliest start time of J (1 mark)

c. Number of activities with zero float (1 mark)

d. Maximum reduction in completion time if A and B are each cut by two days (1 mark)

e. Reductions for each activity that give the earliest finish within a $15 000 budget (1 mark)

Question 12 (5 marks)

2024 VCAA Exam 2 · Question 15 · open in the official exam PDF (p. 25) ↗

An 11-activity supermarket upgrade network: critical path, float, a dummy, new completion time and crashing cost

a. Critical path (1 mark)

b. Activity that can be delayed the longest (1 mark)

c. Draw the dummy activity needed after the change in order (1 mark)

d. New minimum completion time (1 mark)

e. Minimum cost to reduce the completion time as much as possible (1 mark)

Question 13 (2 marks)

2025 VCAA Exam 2 · Question 17 · open in the official exam PDF (p. 25) ↗

Inspecting every walkway between eight stations, starting and ending at A

a. Why some walkways must be repeated, using vertex degrees (1 mark)

b. Minimum distance for the inspection (1 mark)

Marking guide

Mark yourself only after you've finished both exams. Tick the boxes you earned, enter your marks and compare with how Victoria went on each question.

Exam 1 · multiple choice

✓	Q	Ans	Why	How Victoria went
☐	1	**C**	Take $\log_{10}$ of each price: 3175, 4999 and 8925 give 3.50, 3.70 and 3.95, which all lie in the tallest class (3.5 to 4.0).	54% correct A 9% · B 11% · C 54% · D 13% · E 13%
☐	2	**A**	Read two points on the drawn line and find the slope (about 1.55) and intercept (about −6.83). B and C use the wrong slope.	60% correct A 60% · B 15% · C 6% · D 7% · E 12%
☐	3	**C**	Compare each point with the line: five points have residuals between −2 and 2.	43% correct A 17% · B 15% · C 43% · D 13% · E 11%
☐	4	**B**	Slope = (9.9 − 32.2) ÷ (14.9 − 8.53) = −3.5.	59% correct A 9% · B 59% · C 18% · D 10% · E 3%
☐	5	**A**	Negative association: less time on social media goes with higher scores. Association alone never shows causation.	59% correct A 59% · B 9% · C 5% · D 22% · E 4%
☐	6	**D**	Transform age to 1/age on CAS and fit height on 1/age: height = 13.04 − 40.22 × (1/age).	60% correct A 7% · B 5% · C 24% · D 60% · E 3%
☐	7	**B**	Order the 23 values by height on the graph and take the 12th: 117.2 s. A common error took the middle year, 2011, instead.	37% correct A 13% · B 37% · C 11% · D 8% · E 30%
☐	8	**D**	Deseasonalised = 1.35 × actual, so index = actual ÷ deseasonalised = 1 ÷ 1.35 ≈ 0.74.	43% correct A 5% · B 39% · C 9% · D 43% · E 4%
☐	9	**C**	IQR = 48 − 5 = 43; upper fence = 48 + 1.5 × 43 = 112.5. The smallest whole number above it is 113.	53% correct A 17% · B 19% · C 53% · D 11%
☐	10	**A**	b = r × s_y ÷ s_x, so r = −1.27 × 8.51 ÷ 19.0 ≈ −0.569 (negative, like the slope).	52% correct A 52% · B 13% · C 13% · D 20%
☐	11	**D**	Fit pairs on $\log_{10}$(year) with CAS: pairs = 303 − 151 × $\log_{10}$(year).	59% correct A 16% · B 8% · C 18% · D 59%
☐	12	**A**	Fit 1/pairs against year and compare actual with predicted values: the first point (year 1) is furthest from the line.	48% correct A 48% · B 22% · C 15% · D 13%
☐	13	**C**	12 monthly indices add to 12, so months 3 and 6 total 2.07. Month 3 = ⅔ × 2.07 = 1.38.	58% correct A 19% · B 16% · C 58% · D 7%
☐	14	**D**	Density = 6 028 460 ÷ 720 ≈ 8373 people per km², and $\log_{10}$ 8373 ≈ 3.92, which lies in column D.	52% correct A 5% · B 13% · C 30% · D 52%
☐	15	**C**	Fit life on doctors² with CAS, then substitute doctors² = 4: about 74.5 years.	61% correct A 14% · B 14% · C 61% · D 10%

☐	16	**B**	Deseasonalised = actual ÷ 1.75 ≈ 0.571 × actual, a reduction of about 43%.	50% correct A 8% · B 50% · C 18% · D 23%
☐	17	**D**	$T_1 = -5$, then $T_2 = 5$.	43% correct A 13% · B 29% · C 10% · D 43% · E 4%
☐	18	**D**	R = 1.00325 means 3.9% p.a. Finance Solver gives N = 300; the balance before the last payment, grown by one month's interest, is $2614.12.	39% correct A 17% · B 19% · C 17% · D 39% · E 8%
☐	19	**D**	First quarter: (19 527.56 + 653.65) ÷ 20 000 = 1.00906, so the effective rate is $(1.00906^4 - 1) \times 100\% \approx 3.67\%$.	37% correct A 22% · B 16% · C 15% · D 37% · E 9%
☐	20	**B**	A perpetuity's value stays at a: Ra − d = a, so R = (a + d) ÷ a.	27% correct A 4% · B 27% · C 10% · D 37% · E 21%
☐	21	**D**	A geometric sequence multiplies by R each time, so d must be 0: R = 2, d = 0.	36% correct A 14% · B 25% · C 24% · D 36%
☐	22	**B**	Compound: $2000 \times 1.011^{12} - 2000 = 280.57$. Simple: 280.57 ÷ (2000 × 3) = 4.68% per annum.	48% correct A 23% · B 48% · C 18% · D 10%
☐	23	**A**	R = 1.002 means 2.4% p.a. After 24 months FV = 21 340.18. Then N = 36, PV = −21 340.18, FV = 30 000: Pmt ≈ −189.55.	46% correct A 46% · B 15% · C 20% · D 17%
☐	24	**B**	The effective rate applies per year: $5000 \times (1.0451^5 - 1) \approx \$1234$.	41% correct A 10% · B 41% · C 17% · D 32%
☐	25	**A**	Column N of T is (0, 1, 0): birds at N never leave. In the long run every bird ends up at N, so none remain at M (or O).	39% correct A 39% · B 11% · C 21% · D 7% · E 21%
☐	26	**E**	Rows: (0, 1), (1, 0), (4, 1).	62% correct A 4% · B 9% · C 15% · D 10% · E 62%
☐	27	**D**	Not every square matrix has an inverse (det = 0). The other three statements are true.	39% correct A 2% · B 12% · C 42% · D 39% · E 5%
☐	28	**C**	Complete T (columns add to 1), then S_1 and S_2: 140 at the oval on Tuesday and 127 on Wednesday. 0.3 × 140 = 42 stayed, and 42 ÷ 127 ≈ 33%. Using 0.3 itself was the common error.	18% correct A 9% · B 54% · C 18% · D 12% · E 5%
☐	29	**B**	det = 4h − 8g = 0 gives g = h ÷ 2.	50% correct A 17% · B 50% · C 17% · D 15%
☐	30	**C**	$G_4 = G_1P^3$ (or apply P three times): the fifth position in week 4 holds Siobhan.	49% correct A 9% · B 14% · C 49% · D 29%
☐	31	**D**	The remaining games are I v J and I v L. Checking all four outcomes with one-step plus two-step dominances, J can never finish fifth.	53% correct A 17% · B 12% · C 18% · D 53%
☐	32	**D**	Rearrange to $S_n = T^{-1}(S_{n+1} - A)$ and step back three days: 4924 on Thursday, 5620 on Sunday, an increase of 696.	33% correct A 24% · B 19% · C 21% · D 33%
☐	33	**E**	Recheck the minimum cut for each change: reversing GF raises the maximum flow by 7, the most of the five.	40% correct A 15% · B 8% · C 21% · D 15% · E 40%

☐	34	**B**	Redraw it as planar. With 7 vertices and 11 edges, Euler gives 7 + f = 11 + 2, so f = 6.	52% correct A 14% · B 52% · C 9% · D 25%
☐	35	**D**	Test each pair of values in the network: only x = 11 and y = 5 gives a shortest path of exactly 34 m.	40% correct A 20% · B 22% · C 16% · D 40%
☐	36	**A**	Only the degree sum (2 × 6 = 12) is certain. A counterexample graph has four odd vertices, no Eulerian trail and no Hamiltonian path.	18% correct A 18% · B 32% · C 33% · D 17%
☐	37	**D**	The initial optimal allocation takes 43 hours. Re-solving with Edgar replacing each worker in turn, the best is 39 hours: 4 hours less.	19% correct A 11% · B 45% · C 23% · D 19%
☐	38	**B**	Two activities in the network have four immediate predecessors. Check each table for that structure and the path lengths leading to them: table B.	32% correct A 35% · B 32% · C 19% · D 13%
☐	39	**A**	By inspection or Dijkstra: 900 + 500 + 400 + 800 + 1100 = 3700 m.	47% correct A 47% · B 47% · C 4% · D 1%
☐	40	**C**	Forward and backward scans give a 44-day project; B has EST 4 and LST 12, so its float is 8 days.	39% correct A 24% · B 19% · C 39% · D 16%

Multiple-choice score: ☐ **/ 40** **(state average 17.9)**

Exam 2 · written response

Question 1 (6 marks) · state average 3.3 / 6 (55%)

2023 VCAA Exam 2 · Question 3

a. Draw the given least squares line on the scatterplot

☐ Straight line through two correctly calculated points across the data range

Answer / notes: Calculate two points far apart, e.g. temperature −10 gives 0.1164 and 25 gives 0.2004, and join them with a ruler.

State average 0.4 / 1 (40%); 40% scored full marks. Mark distribution: 0: 60% · 1: 40%

Your mark: ☐ **/ 1**

b. Correlation coefficient from $r^2 = 0.7212$, to three decimal places

☐ r = 0.849

Answer / notes: $r = +\sqrt{0.7212} = 0.849$ (positive slope). Some squared r^2 instead.

State average 0.7 / 1 (70%); 69% scored full marks. Mark distribution: 0: 31% · 1: 69%

Your mark: ☐ **/ 1**

c. Describe the association: strength, direction and form

☐ strong, positive, linear

Answer / notes: Strong, positive, linear. “Moderate” came from using r^2 instead of r.

State average 0.6 / 1 (60%); 58% scored full marks. Mark distribution: 0: 42% · 1: 58%

Your mark: ☐ **/ 1**

d. Interpret the intercept

☐ When temperature is 0 °C, predicted consumption is 0.1404 litres per person

Answer / notes: When the average monthly temperature is 0 °C, predicted consumption is 0.1404 L/person. Many interpreted the slope instead.

State average 0.4 / 1 (40%); 43% scored full marks. Mark distribution: 0: 57% · 1: 43%

Your mark: ☐ **/ 1**

e. Predicted consumption at −6 °C

☐ 0.126 L/person

Answer / notes: 0.1404 + 0.0024 × (−6) = 0.126. Rounding to 0.13 lost the mark.

State average 0.8 / 1 (80%); 75% scored full marks. Mark distribution: 0: 25% · 1: 75%

Your mark: ____ **/ 1**

f. Is that prediction interpolation or extrapolation?

☐ extrapolation

Answer / notes: Extrapolation: −6 °C is outside the range of the temperature data, even though it fits on the axes.

State average 0.4 / 1 (40%); 41% scored full marks. Mark distribution: 0: 59% · 1: 41%

Your mark: ____ **/ 1**

Question 2 (10 marks) · state average 5.8 / 10 (58%)

2024 VCAA Exam 2 · Question 3

a. Response variable in Wbest = 0.300 + 0.860 × Wgold

☐ Wbest

Answer / notes: Wbest.

State average 0.9 / 1 (90%); 90% scored full marks. Mark distribution: 0: 10% · 1: 90%

Your mark: ____ **/ 1**

b. Draw the least squares line on the scatterplot

☐ Accurate straight line through two calculated points

Answer / notes: Calculate two points (e.g. Wgold 1.92 → 1.951, 2.06 → 2.072) and rule the line; lines drawn by eye lost the mark.

State average 0.5 / 1 (50%); 51% scored full marks. Mark distribution: 0: 49% · 1: 51%

Your mark: ____ **/ 1**

c. Coefficient of determination as a percentage, to one decimal place, from r = 0.9318

☐ 86.8%

Answer / notes: $0.9318^2 \approx 0.868 = 86.8\%$.

State average 0.6 / 1 (60%); 57% scored full marks. Mark distribution: 0: 43% · 1: 57%

Your mark: ____ **/ 1**

d. Strength and direction of the association

☐ strong, positive

Answer / notes: Strong, positive.

State average 0.8 / 1 (80%); 78% scored full marks. Mark distribution: 0: 22% · 1: 78%

Your mark: ____ **/ 1**

e. Interpret the slope

☐ On average, Wbest increases by 0.86 m for each 1 m increase in Wgold

Answer / notes: Refer to the change in both variables.

State average 0.6 / 1 (60%); 54% scored full marks. Mark distribution: 0: 46% · 1: 54%

Your mark: ____ **/ 1**

f. Show the residual for 1984 (Wgold 2.02, Wbest 2.07) is 0.0328

- [ ] Predicted = 0.30 + 0.86 × 2.02 = 2.0372
- [ ] Residual = 2.07 − 2.0372 = 0.0328

Answer / notes: Both the prediction and the subtraction were needed.

State average 1.0 / 2 (50%); 42% scored full marks. Mark distribution: 0: 46% · 1: 12% · 2: 42%

Your mark: ☐ **/ 2**

g.i. Add that residual to the residual plot

- [ ] Cross at (2.02, 0.0328)

Answer / notes: Plot (2.02, 0.0328). Some didn't connect the scatterplot to the residual plot.

State average 0.5 / 1 (50%); 47% scored full marks. Mark distribution: 0: 53% · 1: 47%

Your mark: ☐ **/ 1**

g.ii. Does the residual plot support a linear model?

- [ ] Yes: no clear pattern in the residuals

Answer / notes: Yes, the residuals are randomly scattered with no pattern. Counting points above and below the line is irrelevant.

State average 0.4 / 1 (40%); 40% scored full marks. Mark distribution: 0: 60% · 1: 40%

Your mark: ☐ **/ 1**

h. Why a prediction for Wgold = 1.90 m isn't reliable

- [ ] Extrapolation: 1.90 is outside the range of the Wgold data

Answer / notes: It's extrapolation, judged on the explanatory variable (Wgold), not the year.

State average 0.5 / 1 (50%); 50% scored full marks. Mark distribution: 0: 50% · 1: 50%

Your mark: ☐ **/ 1**

Question 3 (8 marks) · state average 4.7 / 8 (59%)

2025 VCAA Exam 2 · Question 4

a. Explanatory variable

- [ ] distance from city centre

Answer / notes: Distance from city centre.

State average 1.0 / 1 (100%); 94% scored full marks. Mark distribution: 0: 6% · 1: 94%

Your mark: ☐ **/ 1**

b. Correlation coefficient from $r^2 = 0.0806$, to three decimal places

- [ ] $r = -0.284$

Answer / notes: $r = -\sqrt{0.0806} = -0.284$. Most gave +0.284, missing the negative slope.

State average 0.2 / 1 (20%); 21% scored full marks. Mark distribution: 0: 79% · 1: 21%

Your mark: ☐ **/ 1**

c. Predicted price for a home in the city centre

- [ ] $1 765 353

Answer / notes: Distance 0 gives the intercept, $1 765 353.

State average 0.8 / 1 (80%); 73% scored full marks. Mark distribution: 0: 27% · 1: 73%

Your mark: ☐ **/ 1**

d. Is a prediction at 2 km interpolation or extrapolation? Explain

- [ ] Extrapolation: 2 km is outside the range of the distance data

Answer / notes: Judge it on the explanatory variable (distance), not the price.

State average 0.3 / 1 (30%); 27% scored full marks. Mark distribution: 0: 73% · 1: 27%

Your mark: ____ **/ 1**

e. Strength and direction of the association

- [ ] weak
- [ ] negative

Answer / notes: Weak (r = −0.284) and negative. One mark each.

State average 1.5 / 2 (75%); 65% scored full marks. Mark distribution: 0: 15% · 1: 20% · 2: 65%

Your mark: ____ **/ 2**

f.i. Show the residual for a home 15.5 km away that sold for $1 250 000 is 27 984

- [ ] Predicted = 1 765 353 − 35 054 × 15.5 = 1 222 016; residual = 1 250 000 − 1 222 016 = 27 984

Answer / notes: Show both steps.

State average 0.4 / 1 (40%); 41% scored full marks. Mark distribution: 0: 59% · 1: 41%

Your mark: ____ **/ 1**

f.ii. Plot that residual on the residual plot

- [ ] X at (15.5, 27 984)

Answer / notes: Take care with the vertical scale.

State average 0.5 / 1 (50%); 45% scored full marks. Mark distribution: 0: 55% · 1: 45%

Your mark: ____ **/ 1**

Question 4 (5 marks) · state average 1.6 / 5 (32%)

2023 VCAA Exam 2 · Question 7

a. Show the interest rate is 7.8% per annum

- [ ] (1.0015 − 1) × 52 × 100 = 7.8%

Answer / notes: (1.0015 − 1) × 52 × 100 = 7.8%. Every step had to be shown; CAS “solve” syntax wasn’t accepted.

State average 0.3 / 1 (30%); 32% scored full marks. Mark distribution: 0: 68% · 1: 32%

Your mark: ____ **/ 1**

b.i. d for interest-only repayments

- [ ] 90

Answer / notes: 0.0015 × 60 000 = 90.

State average 0.5 / 1 (50%); 47% scored full marks. Mark distribution: 0: 53% · 1: 47%

Your mark: ____ **/ 1**

b.ii. d to repay the loan fully in five years

- [ ] 278.86

Answer / notes: N = 260, I% = 7.8, PV = 60 000, FV = 0, PpY = CpY = 52: Pmt = −278.86.

State average 0.4 / 1 (40%); 40% scored full marks. Mark distribution: 0: 60% · 1: 40%

Your mark: ____ **/ 1**

c. New d to repay the loan in three more years after a year of $300 payments

- [ ] 350.01

Answer / notes: Find the balance after 52 payments of 300, then solve for Pmt with N = 156: $350.01.

State average 0.3 / 1 (30%); 32% scored full marks. Mark distribution: 0: 68% · 1: 32%

Your mark: ____ **/ 1**

d. Value of d that makes the recurrence relation geometric

- [ ] 0

Answer / notes: d = 0 (multiply only). Only 11% got this.

State average 0.1 / 1 (10%); 11% scored full marks. Mark distribution: 0: 89% · 1: 11%

Your mark: ____ **/ 1**

Question 5 (2 marks) · state average 0.6 / 2 (30%)

2024 VCAA Exam 2 · Question 8

Total cost of the loan to the nearest cent, and the number of payments

- [ ] 288 payments
- [ ] Total cost $884 633.62

Answer / notes: N = 288. After 288 payments FV ≈ −4.18, so $4.18 is still owed and the last payment is $4.18 more. Total = 288 × 3071.63 + 4.18 = $884 633.62. Using 289 payments gave a final payment of about $4, which isn't "slightly different".

State average 0.6 / 2 (30%); 13% scored full marks. Mark distribution: 0: 55% · 1: 31% · 2: 13%

Your mark: ____ **/ 2**

Question 6 (3 marks) · state average 1.1 / 3 (37%)

2025 VCAA Exam 2 · Question 9

a. Weekly interest rate if the balance never changes

- [ ] 0.15%

Answer / notes: 75 ÷ 50 000 = 0.15% per week.

State average 0.4 / 1 (40%); 37% scored full marks. Mark distribution: 0: 63% · 1: 37%

Your mark: ____ **/ 1**

b.i. Annual rate (two decimal places) if the balance after a year is $49 565.34

- [ ] 6.96%

Answer / notes: N = 52, PV = 50 000, Pmt = −75, FV = −49 565.34, PpY = CpY = 52: I% = 6.96.

State average 0.4 / 1 (40%); 40% scored full marks. Mark distribution: 0: 60% · 1: 40%

Your mark: ____ **/ 1**

b.ii. Value of R to four decimal places

- [ ] 1.0013

Answer / notes: R = 1 + 6.96 ÷ (100 × 52) = 1.0013.

State average 0.3 / 1 (30%); 30% scored full marks. Mark distribution: 0: 70% · 1: 30%

Your mark: ____ **/ 1**

Question 7 (2 marks) · state average 0.4 / 2 (20%)

2025 VCAA Exam 2 · Question 10

Complete D_0 and the payment term in $D_{n+1} = 1.016D_n + \ldots$

- [ ] Quarterly payment 22 126.27 (N = 40)
- [ ] $D_0 = 376\,159.43$ and $D_{n+1} = 1.016D_n - 22\,126.27$

Answer / notes: Solve for Pmt with N = 40, then FV after 20 quarters. Some found the payment but used the wrong D_0 or sign.

State average 0.4 / 2 (20%); 11% scored full marks. Mark distribution: 0: 77% · 1: 12% · 2: 11%

Your mark: ____ **/ 2**

Question 8 (4 marks) · state average 1 / 4 (25%)

2024 VCAA Exam 2 · Question 12

a. Percentage decrease in foremen from 2023 to 2025

- [ ] 14%

Answer / notes: Build T from the diagram, find S_2, and compare foremen with 50: a 14% decrease.

State average 0.3 / 1 (30%); 25% scored full marks. Mark distribution: 0: 75% · 1: 25%

Your mark: ____ **/ 1**

b. Total staff on site in the long term

- [ ] 0

Answer / notes: Leaving is absorbing (1.0 on L), so eventually everyone leaves.

State average 0.3 / 1 (30%); 24% scored full marks. Mark distribution: 0: 76% · 1: 24%

Your mark: ____ **/ 1**

c. Increase in staff on site from 2023 to 2024 in the new model

- [ ] 101

Answer / notes: Find V_1 and subtract those in L: 491 on site vs 390 is 101 more. 190 forgot the workers who left.

State average 0.1 / 1 (10%); 12% scored full marks. Mark distribution: 0: 88% · 1: 12%

Your mark: ____ **/ 1**

d. First year the number of foremen exceeds 200

- [ ] 2027

Answer / notes: Generate V_n until foremen first exceed 200 (n = 4, i.e. 2027; “the fourth year” was also accepted).

State average 0.3 / 1 (30%); 26% scored full marks. Mark distribution: 0: 74% · 1: 26%

Your mark: ____ **/ 1**

Question 9 (4 marks) · state average 1.2 / 4 (30%)

2025 VCAA Exam 2 · Question 13

a. What the zeros on the leading diagonal mean

- [ ] No child does the same activity in two consecutive weeks

Answer / notes: Nobody stays in the same activity from one week to the next.

State average 0.7 / 1 (70%); 67% scored full marks. Mark distribution: 0: 33% · 1: 67%

Your mark: ____ **/ 1**

b.i. Expected percentage in cooking in week 10 if all 27 start in cooking

- [ ] 37.4%

Answer / notes: $S_{10} = K^9S_1$; cooking ÷ 27 ≈ 37.4%. Don’t round the expected number mid-way.

State average 0.1 / 1 (10%); 9% scored full marks. Mark distribution: 0: 91% · 1: 9%

Your mark: ____ **/ 1**

b.ii. Expected number in gardening in week 3 who move to music in week 4

- [ ] Gardening in week 3 = 7.776
- [ ] 7.776 × 0.24 ≈ 2

Answer / notes: 2 children. Show working for 2-mark questions.

State average 0.4 / 2 (20%); 17% scored full marks. Mark distribution: 0: 77% · 1: 7% · 2: 17%

Your mark: ☐ **/ 2**

Question 10 (3 marks) · state average 1.1 / 3 (37%)

2025 VCAA Exam 2 · Question 14

a. Activities always held at the same time

- [ ] gardening, lunch, music

Answer / notes: Their rows of P have the 1 on the leading diagonal.

State average 0.7 / 1 (70%); 65% scored full marks. Mark distribution: 0: 35% · 1: 65%

Your mark: ☐ **/ 1**

b. Order of activities on day three

- [ ] drama, cooking, gardening, lunch, music, sport, reading

Answer / notes: Apply P twice (P^2X).

State average 0.3 / 1 (30%); 27% scored full marks. Mark distribution: 0: 73% · 1: 27%

Your mark: ☐ **/ 1**

c. What $P^4 = I$ means over the 40-day program

- [ ] The order repeats every four days (10 cycles in 40 days)

Answer / notes: After four days the timetable is back to day one's order. Few understood the identity matrix here.

State average 0.1 / 1 (10%); 11% scored full marks. Mark distribution: 0: 89% · 1: 11%

Your mark: ☐ **/ 1**

Question 11 (5 marks) · state average 2.1 / 5 (42%)

2023 VCAA Exam 2 · Question 14

a. Immediate predecessors of activity I

- [ ] C, G

Answer / notes: C and G (G via the dummy).

State average 0.7 / 1 (70%); 73% scored full marks. Mark distribution: 0: 27% · 1: 73%

Your mark: ☐ **/ 1**

b. Earliest start time of J

- [ ] 16

Answer / notes: max(4 + 10, 9 + 7) = 16 days.

State average 0.6 / 1 (60%); 59% scored full marks. Mark distribution: 0: 41% · 1: 59%

Your mark: ☐ **/ 1**

c. Number of activities with zero float

- [ ] 5

Answer / notes: A, C, I, K and L. Some counted the dummy.

State average 0.4 / 1 (40%); 35% scored full marks. Mark distribution: 0: 65% · 1: 35%

Your mark: ☐ **/ 1**

d. Maximum reduction in completion time if A and B are each cut by two days

☐ 2

Answer / notes: The project drops from 24 to 22 days: a reduction of 2. Some gave 22.

State average 0.3 / 1 (30%); 33% scored full marks. Mark distribution: 0: 67% · 1: 33%

Your mark: ☐ **/ 1**

e. Reductions for each activity that give the earliest finish within a $15 000 budget

☐ A 2, B 2, F 0, H 2, I 2, K 1

Answer / notes: A 2, B 2, F 0, H 2, I 2, K 1. Only 9% managed this.

State average 0.1 / 1 (10%); 9% scored full marks. Mark distribution: 0: 91% · 1: 9%

Your mark: ☐ **/ 1**

Question 12 (5 marks) · state average 1.6 / 5 (32%)

2024 VCAA Exam 2 · Question 15

a. Critical path

☐ A – C – H – J

Answer / notes: A–C–H–J (29 weeks).

State average 0.7 / 1 (70%); 65% scored full marks. Mark distribution: 0: 35% · 1: 65%

Your mark: ☐ **/ 1**

b. Activity that can be delayed the longest

☐ E

Answer / notes: E has the largest float. Extra information was fine only if it was correct.

State average 0.4 / 1 (40%); 34% scored full marks. Mark distribution: 0: 66% · 1: 34%

Your mark: ☐ **/ 1**

c. Draw the dummy activity needed after the change in order

☐ Directed, labelled dummy edge in the correct place

Answer / notes: The dummy needs an arrow and a label; dashed or solid lines were fine.

State average 0.1 / 1 (10%); 10% scored full marks. Mark distribution: 0: 90% · 1: 10%

Your mark: ☐ **/ 1**

d. New minimum completion time

☐ 30 weeks

Answer / notes: 30 weeks.

State average 0.3 / 1 (30%); 27% scored full marks. Mark distribution: 0: 73% · 1: 27%

Your mark: ☐ **/ 1**

e. Minimum cost to reduce the completion time as much as possible

☐ $50 000

Answer / notes: Reduce critical activities (at most 2 weeks each, not below 7 weeks) at $10 000 per week until no further reduction is possible: $50 000. Only 7% got this.

State average 0.1 / 1 (10%); 7% scored full marks. Mark distribution: 0: 93% · 1: 7%

Your mark: ☐ **/ 1**

Question 13 (2 marks) · state average 0.7 / 2 (35%)

2025 VCAA Exam 2 · Question 17

a. Why some walkways must be repeated, using vertex degrees

- [ ] C and E have odd degree, but a circuit using every edge once needs all even degrees

Answer / notes: An Eulerian circuit needs every vertex even.

State average 0.4 / 1 (40%); 40% scored full marks. Mark distribution: 0: 60% · 1: 40%

Your mark: ____ **/ 1**

b. Minimum distance for the inspection

- [ ] 196 m

Answer / notes: Total of all walkways plus the shortest route between the odd vertices C and E (repeated): 196 m.

State average 0.3 / 1 (30%); 27% scored full marks. Mark distribution: 0: 73% · 1: 27%

Your mark: ____ **/ 1**

Exam 1: ____ **/ 40** **Exam 2:** ____ **/ 59** **Total:** ____ **/ 99**

State average on these exact questions: Exam 1 17.9 / 40, Exam 2 25.2 / 59, total 43.2 / 99 (44%). The notes paraphrase the VCAA examiners' reports; read the full reports for sample answers.